



CSE350: Digital Electronics and Pulse Techniques Course Policy [Spring 2026]

A. Course General Information:

Course Code:	CSE350
Course Title:	Digital Electronics and Pulse Techniques Digital Electronics and Pulse Techniques Laboratory
Credit Hours (Theory+Lab):	3 + 0
Contact Hours (Theory+Lab):	3 + 3
Category:	Program Core
Type:	Required, Engineering, Lecture + Laboratory
Prerequisites:	CSE251: Electronic Devices and Circuits
Co-requisites:	None

B. Course Catalog Description(Content):

This course provides an introduction to basic building blocks of modern digital systems like diode logic gates, transistor switches, transistor gates, MOS gates, Logic families: TTL, ECL, IIL and CMOS logic with operation details. Propagation delay, product and noise immunity. Open collector and High impedance gates. Electronic circuits for flip flops, counters and register, memory systems. PLA's (A/D, D/A converters with applications, S/H circuits) LED, LCD and optically coupled oscillators. Non-linear applications of OPAMPs. Analog switches. Linear wave shaping: diode wave shaping techniques, clipping and clamping circuits, comparator circuits, switching circuits. Pulse transformers, pulse transmission. Pulse generation: mono-stable, bi-stable and stable multi-vibrations, Timing circuits. Simple voltage sweeps, linear circuit sweeps. Schmitt trigger, blocking oscillators and time base circuit. The course includes a compulsory 3 hour laboratory work each week.

C. Course Objective:

The objectives of this course are to

- a. Familiarization with different logic families of circuits
- b. Teach students how to use basic electronics to realize combinational digital circuits
- c. Teach students how to build simple memory and computational unit for digital system
- d. Expose students in the pulse wave shaping and oscillators using different electronic circuit elements
- e. Introduce students to design pulse generation in different ways and means of transmission

D. Course Outcomes(COs):

Upon successful completion of this course, students will be able to

Sl.	CO Description	Weightage (%)
CO1	Identify and investigate electronic circuit level implementation of a combinational logic function for different logic families	35%
CO2	Calculate the performance specification of saturated and unsaturated logic families	25%
CO3	Develop the concepts of continuous signal generation and discretization using Operational amplifier-based circuits	20%
CO4	Operate laboratory equipment, build, and troubleshoot simple combinational and sequential electronic circuits	15%
CO5	Communicate with clear and well-structured technical information through reports.	5%
CO6	Demonstrate the ability to work effectively as an individual and as a team member.	5%

E. Mapping of CO-PO Taxonomy Domain & Level-Delivery-Assessment Tool:

Sl.	CO Description	POs	Bloom's taxonomy domain/level	Delivery methods and activities	Assessment tools
CO1	Identify and investigate electronic circuit level implementation of a combinational logic function for different logic families	PO1	Cognitive/ Analyze	Lectures, notes, Assignments, lab class	Quiz, Exam, Assignments
CO2	Calculate the performance specification of saturated and unsaturated logic families	PO2	Cognitive/ Apply	Lectures, notes, Assignments, lab class	Quiz, Exam, Assignments
CO3	Develop the concepts of continuous signal generation and discretization using Operational amplifier-based circuits	PO3	Cognitive/ Create	Lectures, notes, Assignments, lab class	Quiz, Exam, Assignments
CO4	Operate laboratory equipment, build, and troubleshoot simple combinational and sequential electronic circuits	PO5	Psychomotor/Manipulation	Lectures, notes, lab class	Lab Project
CO5	Communicate with clear and well-structured technical information through reports.	PO10	Psychomotor/Precision	Lab Class	Lab Reports
CO6	Demonstrate the ability to work effectively as an individual and as a team member.	PO9	Affective/Organization	Discussion, Group Work	Lab Performance

F. Course Materials:

i. Text & Reference Books

Sl.	Title	Author(s)	Publication Year	Edition	Publisher / ISBN
1	<i>Microelectronic Circuits Analysis and Design</i>	Donald A. Neamen	2008	3 rd ed.	McGraw-Hill ISBN-13: 978-0-07-352958-5
2	<i>Integrated Electronics: Analog and Digital Circuits and Systems</i>	Jacob Millman, Christos C. Halkias	2010	2 nd ed.	McGraw-Hill ISBN-13: 9780070151420

ii. Other Materials:

- a. All in One Text: [ZAZ](#)
- b. Lecture Slides: [ZAZ](#), [SDU](#)
- c. Lecture Videos: [Shahnewaz Ahmed](#), [Naziul Haque Arnab](#)
- d. [Lab Sheets](#)

G. Lesson Plan:

i. Theory(2 lectures/week, 1 hr 20 mins each)

No	Topic	Lectures	Related CO
1	ADC: Introduction, Flash ADC, Dual Slope ADC	1-3	CO3
2	DAC: Binary Weighted Resistors DAC, R2R Ladder DAC	4-5	CO3
3	Schmitt Trigger: Inverting & Non-inverting [with & without reference voltage]	6-7	CO3
4	Signal Generators: Square & Triangular Wave Generator	8-10	CO3
Mid Term			
5	Diode Logic (DL): OR & AND gates [Basic Operation, Power Dissipation]	11	CO1
6	Resistor Transistor Logic (RTL): NOT & NOR gates [Basic Operation, Power Dissipation, Noise Margin, Fanout]	12-14	CO2
7	Diode Transistor Logic (DTL): NAND gate [Basic Operation, Power Dissipation, Fanout]	15	CO2
8	Transistor Transistor Logic (TTL): NAND gate [Basic Operation, Power Dissipation]	16	CO1
9	NMOS Logic: NOT & NOR gates (with resistive, enhancement type & depletion type loads) [Basic Operation, Power Dissipation]	17-19	CO1
10	CMOS Logic Implementation	20	CO1
Final Exam			

H. Assessment Tools:

Assessment Tools	Weightage(%)
Attendance	2
Assignment	5
Quiz	18
Mid Term Exam	25
Final Exam	25
Lab Work	25

I. CO Assessment Plan:

Assessment Tools	Course Outcomes					
	CO1	CO2	CO3	CO4	CO5	CO6
Assignment	✓	✓	✓			
Quiz	✓	✓	✓			
Mid Term Exam			✓			
Final Exam	✓	✓				
Lab Work				✓	✓	✓

J. CO Attainment Policy:

As per the Department of CSE Course Outcome Attainment Policy

K. Attendance Policy:

A **70%** attendance in theory classes and a **90%** in lab classes are mandatory to be *eligible for the **Final Exam***. Two *Lates* will result in one *Absent* in both cases.

Percentage of Attendance (%)	Marks
90+	2
85-89	1.75
81-84	1.5
75-79	1.25
70-74	1
<70	0

L. Grading Policy:

As per BRAC University grading policy

M. Course Coordinators:

1. **Theory:** [Ihzaz Abrar Sadik \[ZAZ\]](#), [Tasneem Mazhar \[TSE\]](#)
2. **Lab:** [Tasneem Mazhar \[TSE\]](#)